A **0800MAT201122102** **Pages: 3**

Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Find the partial differential equation by eliminating arbitrary functions f and g from z = f(x) + g(y)	(3)
2	Solve $\frac{\partial^2 z}{\partial x^2}$ = xy	(3)
3	Write the three possible solutions of one dimensional wave equation.	(3)
4	Write any two assumptions used in the derivation of one dimensional heat equation.	(3)
5	Test the continuity at $z = 0$ of $f(z) = \begin{cases} \frac{Im(Z)}{\lvert Z \rvert}, z \neq 0 \\ 0, \quad z = 0 \end{cases}$	(3)
6	Check whether f (z) = $\bar{z}$ is an analytic function?	(3)
7	Evaluate $\oint_c \frac{e^z}{z-5}\,dz$, where c is the circle $\lvert z \rvert$= 4	(3)
8	Find the Taylor series expansion of e^z about z = π.	(3)
9	Give example of (a) removable singularity (b) pole (c) essential singularity	(3)
10	Find the Laurent series expansions of $\frac{1}{z(z-1)}$ about $z = 0$	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11	(a)	Solve $y^2 p - xyq = xz$	7
	(b)	Solve by the method of separation of variables $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ =0, u(x, 0) = $4e^{-3x}$	7
12	(a)	Find the complete integral of $px + qy = pq$ using Charpit's method.	7
	(b)	Form the partial differential equation corresponding to family of spheres with center on z-axis and radius a.	7

Module 2

13 (a) Solve the boundary value problem described by u_{tt}-c^2u_{xx}=0, $0 \leq x \leq \ell$, t≥0 7

u(0, t) = u(ℓ,t)=0, t≥0, u(x, 0) = 10 sin$\left(\frac{\pi x}{\ell}\right)$, $\frac{\partial u}{\partial t}$(x, 0) = 0

(b) Find the temperature $u(x,t)$ in a homogenous bar heat conducting material of length l whose ends kept at 0°c and whose initial temperature is given by $u(x,0) = lx - x^2$. 7

14 (a) Derive one dimensional wave equation. 7

(b) The ends A and B of a rod 10 cm in length are kept at temperatures 0°C and 100°C until the steady state condition prevails. If B is Suddenly reduced to 0°C and kept so . Find the temperature distribution in the rod at time t. 7

Module 3

15 (a) Show that an analytic function $f(z) = u + iv$ is constant if its modulus is constant. 7

(b) Find the image of $1 \leq |z| \leq 2$, $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$ under the mapping $w = z^2$ 7

16 (a) Verify whether $u = x^3 - 3xy^2$ is harmonic and find its conjugate harmonic function v. 7

(b) Find the image of the region between real axis and a line parallel to real axis at y = $\frac{\pi}{2}$ under the mapping W= e^z 7

Module 4

17 (a) Evaluate $\int_C |z|^2\, dz$ where C is the circle |z| = 2. 7

(b) Evaluate $\int_C \frac{z^2+2}{(z-3)^2} dz$ where C is the circle |z| = 4 using Cauchy's integral formula 7

18 (a) Evaluate $\oint_c \frac{e^z}{(z-1)(z-4)}$ dz, where c is $|z|$ = 2 using Cauchy's integral formula 7

(b) Evaluate $\int \frac{3z^2+7z}{z+1}$ dz over (a) $|z|$ = 1.5 (b) $|z+i|$ = 1 7

Module 5

19 (a) Find the Laurent series expansion for f(z) = $\frac{1}{(z-1)(z-2)}$ valid in 7

(a) $1 < |z| < 2$ (b)) $|z|$ >2

7

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$

20 (a) Evaluate $\int_{-\infty}^{\infty} \frac{x^2+2}{(x^2+1)(x^2+4)} dx$ 7

(b) Using residue theorem evaluate $\oint_c \frac{z+1}{z^4-2z^3}$ dz ,where c is $|z| = \frac{1}{2}$ 7

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0800MAT201122102

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

3rd SEMESTER B.TECH DEGREE EXAM DEC 2021 (2019 SCHEME)

COURSE CODE : MAT 201

COURSE NAME : PARTIAL DIFFERENTIAL EQUATIONS AND COMPLEX ANALYSIS

PART A

1. Find the partial differential equation by eliminating arbitrary functions f and g from $z = f(x) + g(y)$.

Sol: $z = f(x) + g(y)$ —— ①

$$\Rightarrow \frac{\partial z}{\partial x} = f'(x) + 0$$

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(f'(x)\right) = 0 \Rightarrow \frac{\partial^2 z}{\partial y \partial x} = 0$$

From ①, $\frac{\partial z}{\partial y} = 0 + g'(y)$

$$\Rightarrow \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = 0 \Rightarrow \frac{\partial^2 z}{\partial x \partial y} = 0$$

Thus, $\frac{\partial^2 z}{\partial x \partial y} = 0$ is the required p.d.e

2. Solve: $\frac{\partial^2 z}{\partial x^2} = xy$

Sol: $\frac{\partial^2 z}{\partial x^2} = xy$; Integrating w.r.t 'x' : $\frac{\partial z}{\partial x} = y\frac{x^2}{2} + f(y)$

" " : $z = y\frac{x^3}{6} + x f(y) + g(y)$

3. Write the 3 possible solutions of one dimensional wave equation.

Sol: $u(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct)$; $k<0$

$u(x,t) = (c_1 x + c_2)(c_3 t + c_4)$; $k=0$

$u(x,t) = (c_1 e^{px} + c_2 e^{-px})(c_3 e^{pct} + c_4 e^{-pct})$; $k>0$

4. Write any two assumptions used in the derivation of one dimensional heat equation.

Sol: (i) The sides of the bar are insulated so that there is no radiation from the sides.

(ii) The temperature at all points of any cross-section is same.

5. Test the continuity at $z=0$ of $f(z) = \begin{cases} \frac{\operatorname{Im} z}{|z|}, & z \neq 0 \\ 0, & z=0 \end{cases}$

Sol: To prove the continuity of $f(z)$ at $z=0$, we must prove the following conditions.

(i) $f(0)$ must exist

(ii) $\lim\limits_{z\to 0} f(z)$ must exist

(iii) $\lim\limits_{z\to 0} f(z) = f(0)$

$z = x+iy$

$Im\,z = y$

Here; $f(z) = \dfrac{Im\,z}{|z|} = \dfrac{y}{\sqrt{x^2+y^2}}$

Let $z\to 0$ along the path $y=mx$.

Now, $\boxed{z\to 0} \Rightarrow x+iy\to 0 \Rightarrow x+imx\to 0 \Rightarrow \boxed{x\to 0}$

$$\lim_{z\to 0} f(z) = \lim_{z\to 0}\frac{y}{\sqrt{x^2+y^2}} = \lim_{x\to 0}\frac{mx}{\sqrt{x^2+m^2x^2}}$$

$$= \lim_{x\to 0}\frac{mx}{\sqrt{x^2(1+m^2)}}$$

$$= \lim_{x\to 0}\frac{mx}{x\sqrt{1+m^2}}$$

$$= \lim_{x\to 0}\frac{m}{\sqrt{1+m^2}}$$

$$= \frac{m}{\sqrt{1+m^2}}$$

Here limit is depending on m. Hence it is not unique. $\therefore \lim\limits_{z\to 0} f(z)$ does not exist $\Rightarrow f(z)$ is not continuous at $z=0$ //

(4)

6. Check whether $f(z)=\bar{z}$ is an analytic function ?

Sol: $f(z)=\bar{z}$

$\Rightarrow u+iv = x-iy \quad \Rightarrow \quad u=x \quad \Rightarrow \frac{\partial u}{\partial x}=1, \quad \frac{\partial u}{\partial y}=0$

$v=-y \quad \Rightarrow \frac{\partial v}{\partial x}=0, \quad \frac{\partial v}{\partial y}=-1$

Here ; $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Since both CR equations are not satisfied, $f(z)=\bar{z}$ is not an analytic function.

7. Evaluate $\oint_C \frac{e^z}{z-5}dz$ where C is the circle $|z|=4$.

Sol: $z-5=0 \Rightarrow z=5$ is the singular point

$z=5$ lies outside $|z|=4$.

$\therefore$ By Cauchy's integral theorem, $\oint_C \frac{e^z}{z-5}dz=0$

8. Find the Taylor series expansion of e^z about $z=\pi$

Sol: $e^z = e^{(z-\pi)+\pi}$

$= e^{z-\pi} \cdot e^{\pi}$

$= e^{\pi}\left[1 + \frac{z-\pi}{1!} + \frac{(z-\pi)^2}{2!} + \cdots\right]$

$e^z = 1+\frac{z}{1!}+\frac{z^2}{2!}+\frac{z^3}{3!}+\cdots$

9. Give example of (a) removable singularity (b) pole (c) essential singularity.

Sol: (a) Removable singularity

Ex: $f(z) = \frac{\sin z}{z}$, $z = 0$ is a singular point.

Now $\frac{\sin z}{z} = \frac{1}{z}\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots\right] = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$

Here there is no 'z terms' with negative power.

(b) Pole

Ex: $f(z) = \frac{e^z}{z}$; $z = 0$ is a singular point.

$$= \frac{1}{z}\left[1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots\right]$$

$$= \frac{1}{z} + \frac{1}{1!} + \frac{z}{2!} + \frac{z^2}{3!} + \cdots$$

Here there are finite no. of 'z term' with negative power. So, $z = 0$ is a pole.

(c) Essential singularity

Ex: $f(z) = e^{1/z}$

$$= 1 + \frac{1/z}{1!} + \frac{(1/z)^2}{2!} + \frac{(1/z)^3}{3!} + \cdots$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2}\cdot\frac{1}{2!} + \frac{1}{z^3}\cdot\frac{1}{3!} + \cdots$$

Here there are infinite no. of 'z term' with negative power. So $z = 0$ is an essential singularity.

10. Find the Laurent series expansion of $\frac{1}{z(z-1)}$ about $z=0$

Sol: $\frac{1}{z(z-1)} = \frac{-1}{z(1-z)} = -\frac{1}{z}\cdot(1-z)^{-1}$

$= -\frac{1}{z}\left[1+z+z^2+z^3+\cdots\right]$

$= -\frac{1}{z} - 1 - z - z^2 - \cdots$

PART B

11. (a) Solve : $y^2p - xyq = xz$

Sol: $Pp + Qq = R \Rightarrow$ A.E : $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Here; $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{xz}$ —(1)

Considering 1st two fractions of (1),

$\frac{dx}{y^2} = \frac{dy}{-xy} \Rightarrow \frac{dx}{y} = \frac{dy}{-x} \Rightarrow -x\,dx = y\,dy$

$\Rightarrow -\frac{x^2}{2} = \frac{y^2}{2} + \frac{c_1}{2}$

$\Rightarrow \boxed{x^2+y^2 = c_1}$ —(2)

Considering last two fractions of (1),

$\frac{dy}{-xy} = \frac{dz}{xz} \Rightarrow \frac{dy}{-y} = \frac{dz}{z}$

$\Rightarrow -\log y = -\log z + \log c$

$\Rightarrow -\log y = \log(zc) \quad \Rightarrow \log y^{-1} = -\log(zc)$

$\Rightarrow \frac{1}{y} = zc$

$\Rightarrow y \cdot z = \frac{1}{c} = c_2 \Rightarrow \boxed{yz = c_2}$ — 3

From 2 & 3, general solution is,

$$\phi\left[x^2+y^2,\ yz\right] = 0$$

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11. b Solve by the method of separation of variables

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0, \quad u(x,0) = 4e^{-3x}$$

Sol: Assume solution as, $u = XY$ —— 1

$$\frac{\partial u}{\partial x} = \frac{dX}{dx}Y, \quad \frac{\partial u}{\partial y} = X\frac{dY}{dy}$$

Substituting in given pde.,

$$\frac{dX}{dx}Y + X\frac{dY}{dy} = 0$$

$\div XY \Rightarrow \frac{1}{X}\frac{dX}{dx} + \frac{1}{Y}\frac{dY}{dy} = 0$

$\Rightarrow \frac{1}{X}\frac{dX}{dx} = \frac{-1}{Y}\frac{dY}{dy} = k$ —— 2

$\frac{1}{X}\frac{dX}{dx} = k \Rightarrow \frac{dX}{X} = k\,dx \Rightarrow \log X = kx + \log c_1$

$\Rightarrow \log\left(\frac{X}{c}\right) = kx$

$\Rightarrow \frac{X}{c} = e^{kx} \Rightarrow X = c_1 e^{kx}$ —— 3

From ② ; $-\frac{1}{Y}\frac{dY}{dy} = k \Rightarrow \frac{dY}{Y} = -k\,dy$

$\Rightarrow \log Y = -ky + \log c_2$

$\Rightarrow \log\left(\frac{Y}{c}\right) = -ky$

$\Rightarrow \frac{Y}{c_2} = e^{-ky} \Rightarrow Y = c_2 e^{-ky}$ —— ④

Sub. ③ & ④ in ①, $u = c_1 e^{kx} \cdot c_2 e^{-ky}$

$\Rightarrow u(x,y) = c_1 c_2 e^{kx-ky}$ —— ⑤

Given: $u(x,0) = 4e^{-3x}$

From ⑤, $u(x,0) = c_1 c_2 e^{kx}$ $\Big\} \Rightarrow c_1 c_2 = 4 \;,\; k = -3$

∴ ⑤ becomes, $u(x,y) = 4\,e^{-3x+3y}$

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12 ⑨ Find the complete integral of $px + qy = pq$ using Charpit's method.

Sol: A.E : $\frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z}$ —— ①

$f = px + qy - pq = 0$ —— ②

$f_p = x - q \qquad f_x = p \qquad f_z = 0$

$f_q = y - p \qquad f_y = q$

$\therefore$ 1 becomes,

$$\frac{dx}{q-x} = \frac{dy}{p-y} = \frac{dz}{-px+pq-qy+pq} = \frac{dp}{p+0} = \frac{dq}{q+0}$$

Hence, $\frac{dp}{p} = \frac{dq}{q}$ from last two fractions

$\Rightarrow \log p = \log q + \log c$

$\Rightarrow \log\left(\frac{p}{q}\right) = \log c \Rightarrow \frac{p}{q} = c \Rightarrow \boxed{p = cq}$

Put $p = cq$ in 2, we get

$cqx + qy = cq^2 \Rightarrow cx + y = cq \Rightarrow \boxed{q = \frac{cx+y}{c}}$

Substitute p and q in $dz = p\,dx + q\,dy$

$$\Rightarrow dz = cq\,dx + \left(\frac{cx+y}{c}\right)dy$$

$$\Rightarrow c\,dz = c^2q\,dx + (cx+y)\,dy$$

$$= c^2\left(\frac{cx+y}{c}\right)dx + (cx+y)\,dy$$

$$c\,dz = (cx+y)(c\,dx+dy)$$

$$= (cx+y)\,d(cx+y)$$

Integrating both sides,

$$cz = \frac{(cx+y)^2}{2} + K$$

12 (b) Form the partial differential equation corresponding to family of spheres with center on z-axis and radius a.

Sol: Writing the equation of sphere mentioned, we've

$x^2+y^2+(z-c)^2=a^2$ — ①

Diff ① w.r.t 'x' : $2x+2(z-c)\frac{\partial z}{\partial x}=0$

$\Rightarrow x+(z-c)p=0$

$\Rightarrow x=-(z-c)p \Rightarrow z-c=\frac{-x}{p}$

Diff ① w.r.t 'y' : $2y+2(z-c)\frac{\partial z}{\partial y}=0$

$\Rightarrow y+(z-c)q=0$

$\Rightarrow y=-(z-c)q \Rightarrow z-c=\frac{-y}{q}$

Equating values of '$z-c$' we get : $-\frac{x}{p}=\frac{-y}{q}$

$\Rightarrow py=qx \Rightarrow \underline{\underline{py-qx=0}}$

13 (a) Solve the boundary value problem described by $u_{tt}-c^2u_{xx}=0$, $0\le x\le l$, $t\geq 0$, $u(0,t)=u(l,t)=0$, $u(x,0)=10\sin\left(\frac{\pi x}{l}\right)$, $\frac{\partial u}{\partial t}(x,0)=0$

Sol: $u_{tt} - c^2 u_{xx} = 0 \Rightarrow \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$ —— (1)

General solution to (1) is,

$$u(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos pct + c_4 \sin pct) \quad \text{—— (2)}$$

Applying $u(0,t)=0$ in (2), we get $c_1 = 0$

Applying $u(l,t)=0$ in (2), we get $p = \frac{n\pi}{l}$

Thus, (2) becomes,

$$u(x,t) = c_2 \sin \frac{n\pi x}{l} \left[c_3 \cos\left(\frac{n\pi ct}{l}\right) + c_4 \sin\left(\frac{n\pi ct}{l}\right) \right]$$

$$= \sin\left(\frac{n\pi x}{l}\right)\left[a_n \cos \frac{n\pi ct}{l} + b_n \sin \frac{n\pi ct}{l} \right]$$

where $a_n = c_2 c_3$
$b_n = c_2 c_4$

By the principle of superposition,

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\left[a_n \cos \frac{n\pi ct}{l} + b_n \sin \frac{n\pi ct}{l} \right] \quad \text{—— (3)}$$

Given: $\frac{\partial u}{\partial t}(x,0) = 0$

Differentiate (3) w.r.t 't'

$$\frac{\partial u(x,t)}{\partial t} = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\left[a_n \cdot \bar{\ } \sin\left(\frac{n\pi ct}{l}\right) \cdot \frac{n\pi c}{l} + b_n \cos\left(\frac{n\pi ct}{l}\right) \cdot \left(\frac{n\pi c}{l}\right) \right]$$

$$\frac{\partial u(x,0)}{\partial t} = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right)\left[0 + b_n \cdot \frac{n\pi c}{l}\right]$$

$$= \sum_{n=1}^{\infty} b_n \cdot \frac{n\pi c}{l} \sin\left(\frac{n\pi x}{l}\right) = 0 \quad \Rightarrow b_n = 0$$

$\therefore$ (3) becomes,

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) \cdot a_n \cdot \cos\left(\frac{n\pi ct}{l}\right) \quad \text{---} (4)$$

Given : $u(x,0) = 10 \sin\left(\frac{\pi x}{l}\right)$

Put $t=0$ in (4) ; $u(x,0) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) a_n$

Equating the values of $u(x,0)$,

$$10 \sin\left(\frac{\pi x}{l}\right) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{l}\right) a_n$$

$$= \sin\left(\frac{\pi x}{l}\right) \cdot a_1 + \sin\left(\frac{2\pi x}{l}\right) a_2 + \cdots$$

Equating the coefficients of $\cancel{a_1, a_2,}$ $\sin\left(\frac{\pi x}{l}\right), \sin\left(\frac{2\pi x}{l}\right), \ldots$

we get $a_1 = 10$, $a_2 = a_3 = a_4 = \cdots = 0$

$\therefore$ Solution (4) becomes,

$$u(x,t) = \sin\left(\frac{\pi x}{l}\right) \cdot a_1 \cos\frac{\pi ct}{l} + 0 + 0 + \cdots$$

$$\Rightarrow u(x,t) = 10 \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{\pi ct}{l}\right)$$ is the required solution.

13 (b) Find the temperature $u(x,t)$ in a homogeneous bar heat conducting material of length l whose ends kept at $0°c$ and whose initial temperature is given by $u(x,0) = lx - x^2$.

Soln :-

We have the heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \longrightarrow (1)$$

Its solution is given by

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-c^2p^2t} \longrightarrow (2)$$

Now $u(0,t) = 0$, $u(l,t) = 0$ as the boundary conditions. $\longrightarrow (3)$

And $u(x,0) = lx - x^2$ as the initial condition. $\longrightarrow (4)$

$u(0,t) = 0$ in (2)

$$\Rightarrow \boxed{c_1 = 0}.$$

$$\therefore u(x,t) = (c_2 \sin px)\, c_3 e^{-c^2p^2t} \longrightarrow (5)$$

$u(l,t) = 0$ in (5)

$$\Rightarrow \quad 0 = (c_2 \sin pl)\, c_3 e^{-c^2p^2t}$$

$$\Rightarrow \quad \sin pl = 0 \Rightarrow \boxed{p = \frac{n\pi}{l}}$$

$$\therefore \quad u(x,t) = \left(c_2 \sin \frac{n\pi x}{l}\right) c_3\, e^{-n^2\pi^2 t}$$

$$= c_2 c_3 \sin \frac{n\pi x}{l}\, e^{-n^2\pi^2 t}$$

$$u(x,t) = b_n \sin \frac{n\pi x}{l}\, e^{-n^2\pi^2 t} \text{ where } b_n = c_2 c_3$$

Adding up all the solutions the general solution is

$$u(x,t) = \sum_1^{\infty} b_n \sin \frac{n\pi x}{l}\, e^{-n^2\pi^2 t} \longrightarrow (6)$$

Applying initial condition

$$u(x,0) = lx - x^2$$

$$(6) \Rightarrow \quad lx - x^2 = \sum_1^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{where}$$

$$b_n = \frac{2}{l} \int_0^{l} (lx - x^2) \sin \frac{n\pi x}{l}\, dx$$

$$= \left[(lx - x^2)\left(\frac{-\cos\left(\frac{n\pi x}{l}\right)}{\frac{n\pi}{l}} \right) - (l - 2x)\left(\frac{-\sin \frac{n\pi x}{l}}{\left(\frac{n\pi}{l}\right)^2} \right) + (-2)\left(\cos \frac{n\pi x}{l} \Big/ \left(\frac{n\pi}{l}\right)^3 \right) \right]_0^{l}$$

$$= \left[0 + (-1)\left(\frac{-\sin\frac{n\pi l}{l}}{\left(\frac{n\pi}{l}\right)^2}\right) - 2\cdot\frac{\cos\left(\frac{n\pi l}{l}\right)}{\left(\frac{n\pi}{l}\right)^3}\right] - \left[\frac{-2}{\left(\frac{n\pi}{l}\right)^3}\right]$$

$$= \frac{-2\cos n\pi}{\left(\frac{n\pi}{l}\right)^3} + \frac{2}{\left(\frac{n\pi}{l}\right)^3}$$

$$= \frac{2l^3}{n^3\pi^3}\left[1 - \cos n\pi\right]$$

$$= \frac{2l^3}{n^3\pi^3}\left[1 - (-1)^n\right]$$

$\therefore$ Soln is

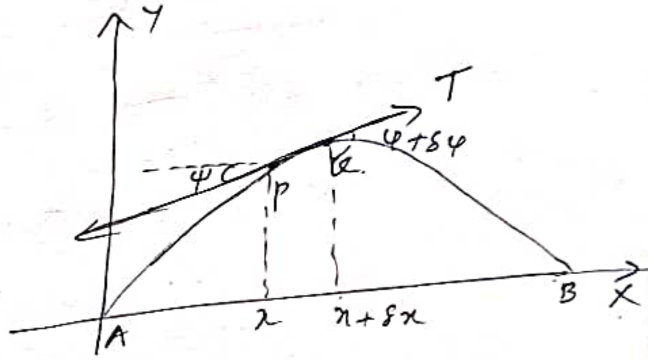
$$u(x,t) = \sum_{1}^{\infty} \frac{2l^3}{n^3\pi^3}\left(1 - (-1)^n\right)\sin\left(\frac{n\pi x}{l}\right)e^{-n^2\pi^2 t}$$

(14)

(a) Derive One-dimensional wave equation.

Soln :-

Consider a tightly stretched elastic string of length 'l' and fixed at ends A and B and subjected to constant tension T.



Let the string be released from rest and allowed to vibrate.

Taking the end A as the origin and AB as the x-axis and AY perpendicular to it as the Y-axis. so that the motion takes place entirely in the xy-plane.

Consider the motion of the element PQ of the string between its points $P(x,y)$ and $Q(x+\delta x, y+\delta y)$ where the tangents make angles ψ and $\psi+\delta\psi$ with the x-axis.

The vertical component of the force acting on this element

$$= T \sin(\varphi + \delta\varphi) - T \sin\varphi.$$

$$= T\left(\sin(\varphi + \delta\varphi) - \sin\varphi\right)$$

$$= T\left(\tan(\varphi + \delta\varphi) - \tan\varphi\right)$$

$$= T\left[\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_{x}\right]$$

Since φ is small $\cos\varphi = 1$ $\Rightarrow \tan\varphi = \sin\varphi$ Similarly $\tan(\varphi+\delta\varphi) = \sin(\varphi+\delta\varphi)$

If 'm' is the mass per unit length of the string, then mass of the element PQ is $m\,\delta x$.

By Newton's 2nd law of motion,

$$(m\,\delta x)\left(\frac{\partial^2 y}{\partial t^2}\right) = T\left[\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_{x}\right]$$

$F = ma$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{m}\left[\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_{x}}{\delta x}\right]$$

Taking limits

$$\lim_{\delta x \to 0} \frac{\partial^2 y}{\partial t^2} = \frac{T}{m} \lim_{\delta x \to 0}\left[\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_{x}}{\delta x}\right]$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{m}\frac{\partial^2 y}{\partial x^2}$$

$$\Rightarrow \boxed{\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}} \text{ where } c^2 = T/m$$

14 (b) The ends A and B of a rod of 10 cm in length are kept at temperatures 0°c and 100°c until steady state conditions prevails. If B is suddenly reduced to 0°c and kept so. Find the temperature distribution in the rod at time 't'..

Soln:-

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \longrightarrow (1)$$

Boundary conditions are

$$\left. u(0,t)=0 \text{ and } u(10,t)=100 \right\} \rightarrow (2)$$

In the steady state condition

$$\frac{\partial u}{\partial t} = 0.$$

$$\therefore (1) \Rightarrow c^2 \frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} = 0 \rightarrow (3)$$

Integrating (3), $\Rightarrow \frac{\partial u}{\partial x} = a \Rightarrow \boxed{u = ax+b}$

$u(0,t)=0 \Rightarrow \boxed{b=0}$

$u(10,t)=0 \Rightarrow 100 = a \times 10 \Rightarrow \boxed{a=10}$

Hence $u(x,0) = 10x \longrightarrow (4)$

As the end B is reduced to 0°c and maintained at 0°c now the boundary conditions are

$$u(0,t) = 0 \text{ and } u(10,t) = 0 \longrightarrow (5)$$

Now we have the solution

$$u(x,t) = (c_1 \cos px + c_2 \sin px)\, c_3 e^{-c^2p^2t} \longrightarrow (6)$$

Now $\underline{u(0,t) = 0}$

$$\Rightarrow \quad 0 = c_1 e^{-c^2p^2t} \Rightarrow \boxed{c_1 = 0}$$

$$\therefore u(x,t) = (c_2 \sin px)\, c_3 e^{-c^2p^2t} \longrightarrow (7)$$

Now $\underline{u(10,t) = 0}$

$$\Rightarrow \quad 0 = (c_2 \sin 10p)\, c_3 e^{-c^2p^2t} \Rightarrow \cancel{p} \sin 10p = 0$$

$$\Rightarrow \boxed{p = \frac{n\pi}{10}}$$

$$\therefore u(x,t) = (c_2 c_3) \sin\left(\frac{n\pi x}{10}\right) \cdot e^{-\frac{c^2n^2\pi^2t}{100}}$$

$$\Rightarrow u(x,t) = b_n \sin\left(\frac{n\pi x}{10}\right) \cdot e^{-\frac{c^2n^2\pi^2t}{100}}$$

where $b_n = c_2 c_3$

So the general soln is

$$u(x,t) = \sum_{1}^{\infty} b_n \sin\left(\frac{n\pi x}{10}\right) \cdot e^{-\frac{c^2 n^2 \pi^2 t}{100}} \longrightarrow (8)$$

Now $u(x,0) = 10x = f(x)$.

$$\therefore (8) \Rightarrow \sum_{1}^{\infty} b_n \sin\left(\frac{n\pi x}{10}\right) = 10x$$

$$\Rightarrow b_n = \frac{2}{10}\int_0^{10} f(x) \sin\left(\frac{n\pi x}{10}\right) dx.$$

$$= \frac{2}{10}\int_0^{10} 10x \sin\left(\frac{n\pi x}{10}\right) dx$$

$$= 2\left[(x)\left(\frac{-\cos\left(\frac{n\pi x}{10}\right)}{\frac{n\pi}{10}}\right) - (1)\left(\frac{-\sin\left(\frac{n\pi x}{10}\right)}{\left(\frac{n\pi}{10}\right)^2}\right)\right]_0^{10}$$

$$= 2\left[\left(10 \cdot \left(\frac{10}{n\pi}\right)\left(-\cos n\pi\right) + 0\right) - 0\right]$$

$$= -2\left(\frac{100}{n\pi} \cdot (-1)^n\right)$$

$$= \frac{200}{n\pi}(-1)^{n+1}$$

$$\therefore u(x,t) = \sum_{1}^{\infty} \frac{200}{n\pi}(-1)^{n+1} \sin\left(\frac{n\pi x}{10}\right) e^{-\frac{c^2 n^2 \pi^2 t}{100}}$$

Module 3

15

(a) Show that an analytic function $f(z) = u + iv$ is constant if its modulus is constant.

Soln:- $f(z) = u + iv$

Given $|f(z)| = \sqrt{u^2+v^2} = \text{constant}$.

$$\Rightarrow u^2 + v^2 = k \longrightarrow (1)$$

Diff. (1) partially w.r.t 'x',

$$2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} = 0$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial v}{\partial x} = 0 \longrightarrow (2)$$

Similarly diff. (1) partially w.r.t 'y',

$$u\frac{\partial u}{\partial y} + v\frac{\partial v}{\partial y} = 0 \longrightarrow (3)$$

Applying CR equations in (3),

$$-u\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial x} = 0 \longrightarrow (4)$$

Now (2) $\times u \Rightarrow u^2\frac{\partial u}{\partial x} + uv\frac{\partial v}{\partial x} = 0 \longrightarrow (5)$

(4) $\times v \Rightarrow -uv\frac{\partial v}{\partial x} + v^2\frac{\partial u}{\partial x} = 0 \longrightarrow (6)$

$(5) + (6) \Rightarrow u^2 \frac{\partial u}{\partial x} + v^2 \frac{\partial u}{\partial x} = 0$

$\Rightarrow (u^2+v^2) \frac{\partial u}{\partial x} = 0$

$\Rightarrow K \frac{\partial u}{\partial x} = 0 \Rightarrow \boxed{\frac{\partial u}{\partial x} = 0}$

Proceeding similarly, $\boxed{\frac{\partial u}{\partial y} = 0}$

$\Rightarrow \frac{\partial u}{\partial x} = 0, \frac{\partial u}{\partial y} = 0 \Rightarrow u =$ constant

$\Rightarrow \frac{\partial v}{\partial y} = 0$ and $-\frac{\partial v}{\partial x} = 0 \Rightarrow v =$ constant [By CR]

$\Rightarrow f(z) = u + iv$ is also constant

15 (b) Find the image of $1 \le |z| \le 2$, $\frac{\pi}{6} \le \theta \le \frac{\pi}{3}$ under $w = z^2$.

Soln :-

Given $w = z^2$.

Let $z = re^{i\theta}$, $w = Re^{i\phi}$.

$w = z^2 \Rightarrow Re^{i\phi} = (re^{i\theta})^2$

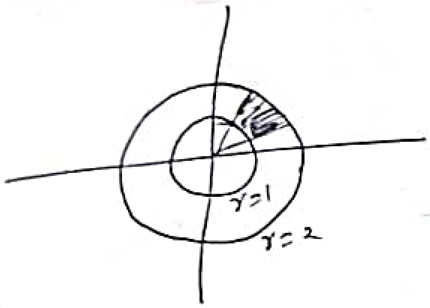
$\Rightarrow \boxed{R = r^2}$ and $\boxed{\phi = 2\theta}$

Given $1 < |z| < 2$

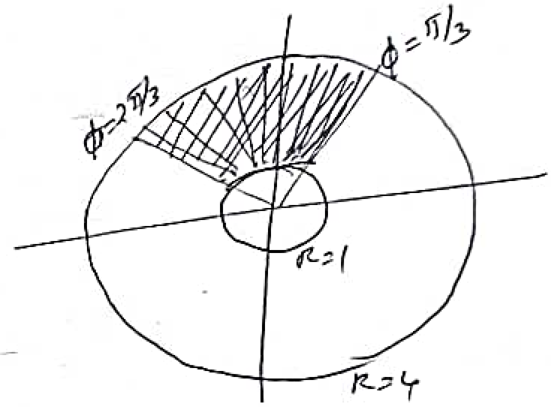
$$\Rightarrow \boxed{1 < r < 2} \Rightarrow 1 < r^2 < 4 \Rightarrow \boxed{1 < R < 4}$$

Also $\frac{\pi}{6} \le \theta \le \pi/3 \Rightarrow \frac{\pi}{6} \le \frac{\phi}{2} \le \pi/3$

$$\Rightarrow \boxed{\frac{\pi}{3} \le \phi \le \frac{2\pi}{3}}$$



Z plane



(16) (a) Verify whether $u = x^2 - 3xy^2$ is harmonic and find its harmonic conjugate function v.

Soln:-

$$u = x^3 - 3xy^2$$

$$\Rightarrow \frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial u}{\partial y} = -6xy$$

By CR, $\frac{\partial v}{\partial y} = 3x^2 - 3y^2 \longrightarrow (1)$

$$\frac{\partial v}{\partial x} = 6xy \longrightarrow (2)$$

Integrating ①, w.r.t y,

$$V = 3x^2y - \frac{3y^3}{3} + g(x).$$

$$\Rightarrow \boxed{V = 3x^2y - y^3 + g(x)} \longrightarrow ③$$

Diff. ③ w.r.t 'x',

$$\frac{\partial V}{\partial x} = 6xy + g'(x). \longrightarrow ④$$

From ② and ④, $g'(x) = 0$

$$\Rightarrow \boxed{g(x) = c}$$

$$\therefore \quad V = 3x^2y - y^3 + c$$

16 b) Find the image of the region between real axis and a line parallel to real axis at $y = \pi/2$ under $w = e^z$.

Soln:-

Let $w = Re^{i\phi}$, $z = x + iy$

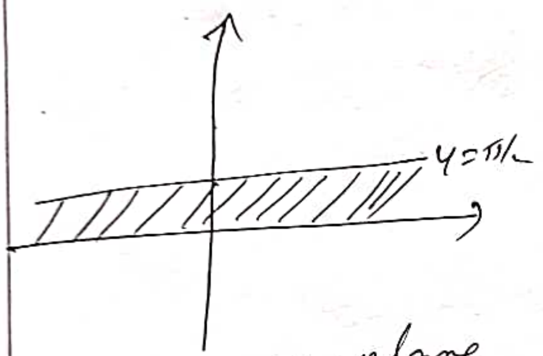
$\therefore w = e^z \Rightarrow Re^{i\phi} = e^{x+iy}$

$$\Rightarrow \boxed{R = e^x} \text{ and } \boxed{\phi = y}$$

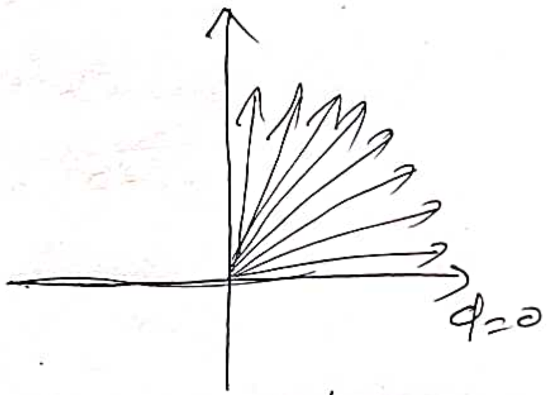
Now the real axis $y=0 \Rightarrow \boxed{\phi = 0}$

and $y = \pi/2 \Rightarrow \boxed{\phi = \pi/2}$

The region between $y=0$ and $y=\pi/2$ in z-plane is transformed to region between $\phi = 0$ and $\phi = \pi/2$ (I^{st} quadrant)



z-plane



w-plane

Module 4

(17) (a) Evaluate $\int_C |z|^2 dz$ where C is $|z|=2$.

Soln:

$Z(t) = x(t) + i\, y(t)$. where $x(t) = 2\cos t$, $y(t) = 2\sin t$

$= 2(\cos t + i \sin t)$

$= 2e^{it}$ where $0 \le t \le 2\pi$

$\Rightarrow \frac{dz}{dt} = 2ie^{it}$

$$\therefore \int_C f(z)\,dz = \int_a^b f(z(t))\,\frac{dz}{dt}\cdot dt$$

$$\Rightarrow \int_C |z|^2\,dz = \int_0^{2\pi} 4\cdot 2ie^{it}\cdot dt$$

$$= 8i\left[\frac{e^{it}}{i}\right]_0^{2\pi}$$

$$= 8\left[e^{i2\pi} - e^0\right]$$

$$= 8\left[(\cos 2\pi + i\sin 2\pi) - 1\right]$$

$$= \underline{\underline{0}}$$

---

17 (b) Evaluate $\int_C \frac{z^2+2}{(z-3)^2}\,dz$ where c is the circle $|z|=4$ using Cauchy's Integral formula.

<u>Soln:-</u>

Here the singularity $z=3$ (of order 2) lies inside $|z|=4$.

We have $f^n(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)\,dz}{(z-z_0)^{n+1}}$

$$\Rightarrow \oint_C \frac{z^2+2}{(z-3)^2}\,dz = \frac{2\pi i}{n!}\, f'(3)$$

$n+1=2$

$\underline{n=1}$

$$= \frac{2\pi i}{1!} \times 6$$

$$= 12\pi i$$

$f(z) = z^2 + 2$

$\Rightarrow f'(z) = 2z$

$\Rightarrow f'(3) = 2 \times 3$

$= 6$

(18) (a) Evaluate $\oint_C \frac{e^z}{(z-1)(z-4)} dz$ where C is $|z| = 2$ using Cauchy's Integral formula.

Soln:-

Singularities are $z = 1$ and $z = 4$ where $z = 1$ lies inside $|z| = 2$ and $z = 4$ lies outside $|z| = 2$

Rearranging the Integral

$$\oint_C \frac{e^z dz}{(z-1)(z-4)} = \oint_C \frac{\frac{e^z}{z-4}}{z-1} dz$$

$$= 2\pi i\, f(1)$$

$$= 2\pi i \times \frac{-e}{3}$$

$$= -\frac{2}{3}\pi e i$$

$f(z) = \frac{e^z}{z-4}$

$f(1) = \frac{e^1}{1-4}$

$= \frac{-e}{3}$

18 (b) Evaluate $\int_C \frac{3z^2+7z}{z+1} dz$ along

(a) $|z| = 1.5$ (b) $|z+i| = 1$

Soln :-

(a) $|z| = 1.5$

singularity is $z = -1$ which lies inside $|z| = 1.5$

$$\therefore \int_C \frac{3z^2+7z}{z+1} dz = 2\pi i \, f(-1)$$
$$= 2\pi i \times -4$$
$$= -8\pi i$$

$f(z) = 3z^2+7z$

$f(-1) = 3-7$
$= -4$

(b) $|z+i| = 1$

When $z = -1$, $|z+i| = |-1+i|$
$= \sqrt{1+1} = \sqrt{2} > 1$

Hence $z = -1$ lies outside $|z+i| = 1$

Hence by Cauchy's theorem

$$\int_C \frac{3z^2+7z}{z+1} dz = 0$$

Module 5

(19) (a) Find the Laurent series expansion of $f(z) = \frac{1}{(z-1)(z-2)}$ valid in

(a) $1 \le |z| < 2$ (b) $|z| > 2$.

Soln:-

$$f(z) = \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$1 = A(z-2) + B(z-1)$$

$z = 1$, $1 = -A \Rightarrow \boxed{A = -1}$

$z = 2$, $1 = B \Rightarrow \boxed{B = 1}$

$$\therefore \frac{1}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{1}{z-2}$$

(a) $1 < |z| < 2 \Rightarrow 1 < |z|, \; |z| < 2$

$$\Rightarrow \frac{1}{|z|} < 1, \; \frac{|z|}{2} < 1$$

$$\frac{1}{(z-1)(z-2)} = \frac{-1}{z(1-1/z)} + \frac{1}{-2(1-\frac{z}{2})}$$

$$= \frac{-1}{z}(1-1/z)^{-1} - \frac{1}{2}(1-z/2)^{-1}$$

$$= \frac{-1}{z}\left[1+\left(\frac{1}{z}\right)+\left(\frac{1}{z}\right)^2+\cdots\right] -$$

$$\frac{1}{2}\left[1+\left(\frac{z}{2}\right)+\left(z/2\right)^2+\cdots\right]$$

(b) $|z| > 2 \Rightarrow \frac{2}{|z|} < 1$

$$\Rightarrow \frac{1}{|z|} < 1$$

$$\frac{1}{(z-1)(z-2)} = \frac{-1}{z(1-1/z)} + \frac{2}{z(1-2/z)}$$

$$= \frac{-1}{z}(1-1/z)^{-1} + \frac{2}{z}(1-2/z)^{-1}$$

$$= \frac{-1}{z}\left[1+(1/z)+(1/z)^2+\cdots\right] +$$

$$\frac{2}{z}\left[1+(2/z)+(2/z)^2+\cdots\right]$$

19(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$

Soln:-

Let $z = e^{i\theta} \Rightarrow d\theta = \frac{dz}{iz}$

Also $\sin\theta = \frac{z - \frac{1}{z}}{2i}$

$\theta : 0$ to $2\pi \Rightarrow C$ is $|z| = 1$

$$\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta} = \int_C \frac{\frac{dz}{iz}}{5-4\left(\frac{z-\frac{1}{z}}{2i}\right)}$$

$$= \int_C \frac{dz}{iz\left[5 - \frac{2}{i}\left(\frac{z^2-1}{z}\right)\right]}$$

$$= \int_C \frac{dz}{iz\left[\frac{5iz - 2z^2 + 2}{iz}\right]}$$

$$= \int_C \frac{dz}{-(2z^2 - 5iz - 2)} \rightarrow (1)$$

Singularities are given by

$$2z^2 - 5iz - 2 = 0$$

$$\Rightarrow z = \frac{5i \pm \sqrt{-25+16}}{4}$$

$$= \frac{5i \pm 3i}{4} = 2i, \; i/2$$

Here $z = \frac{i}{2}$ lies inside $|z|=1$

$$\operatorname*{Res}_{z=i/2} f(z) = \lim_{z \to i/2} (z - i/2) \frac{-1}{2(z-2i)(z-i/2)}$$

$$= \frac{-1}{2 \times \left(\frac{i}{2} - 2i\right)}$$

$$= \frac{-1}{2\left(\frac{i-4i}{2}\right)} = \frac{-1}{-3i} = \frac{1}{3i}$$

Hence $\displaystyle\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta} = 2\pi i \left[\operatorname*{Res}_{z=i/2} f(z)\right]$

$$= 2\pi i \times \frac{1}{3i}$$

$$= \frac{2\pi}{3}$$

20 (a) Evaluate $\int_{-\infty}^{\infty} \frac{x^2+2}{(x^2+1)(x^2+4)} dx.$

Soln :-

Consider $\int_C \frac{z^2+2}{(z^2+1)(z^2+4)} dz$

Singularities are given by

$$(z^2+1)(z^2+4)=0$$

$\Rightarrow z = \pm i, \pm 2i$

$\Rightarrow z = i, z = 2i$ lies in the upper half plane

$$\operatorname*{Res}_{z=i} f(z) = \lim_{z \to i} (z-i) \frac{(z^2+2)}{(z+i)(z-i)(z^2+4)}$$

$$= \frac{-1+2}{(2i)(3)} = \underline{\frac{1}{6i}}$$

$$\operatorname*{Res}_{z=2i} f(z) = \lim_{z\to 2i} \frac{(z-2i)(z^2+2)}{(z^2+1)(z-2i)(z+2i)}$$

$$= \frac{-4+2}{(-4+1)(4i)} = \frac{-2}{-3\times 4i}$$

$$= \frac{2}{12i} = \frac{1}{6i}$$

Hence by ~~joining~~ Cauchy's Residue Theorem,

$$\int_{-\infty}^{\infty} \frac{(x^2+2)\,dx}{(x^2+1)(x^2+4)} = 2\pi i \left[\operatorname*{Res}_{z=i} f(z) + \operatorname*{Res}_{z=2i} f(z)\right]$$

$$= 2\pi i \left[\frac{1}{6i} + \frac{1}{6i}\right]$$

$$= 2\pi i \left[\frac{2}{6i}\right]$$

$$= \frac{2\pi}{3}$$

---

20 (b) Using residue theorem evaluate

$\oint_C \frac{z+1}{z^4-2z^3}\,dz$ where C is $|z| = \frac{1}{2}$.

Soln:-

$$\oint_C \frac{z+1}{z^4-2z^3}\,dz = \oint_C \frac{(z+1)\,dz}{z^3(z-2)}$$

Singularities are given by $z^3(z-2)=0$

$\Rightarrow z=0$ of order 3

$z=2$ of order 1

Here $z=0$ lies inside $|z|=\frac{1}{2}$ and

$z=2$ lies outside $|z|=\frac{1}{2}$.

$$\therefore \underset{z=0}{\text{Res}}\, f(z) = \frac{1}{2!}\lim_{z\to 0}\frac{d^2}{dz^2}\left[(z-0)^3\cdot\frac{z+1}{z^3(z-2)}\right]$$

$$= \frac{1}{2}\lim_{z\to 0}\frac{d^2}{dz^2}\left(\frac{z+1}{z-2}\right)$$

$$= \frac{1}{2}\lim_{z\to 0}\frac{d}{dz}\left(\frac{(z-2)\cdot 1-(z+1)\cdot 1}{(z-2)^2}\right)$$

$$= \frac{1}{2}\lim_{z\to 0}\frac{d}{dz}\frac{-3}{(z-2)^2}$$

$$= \frac{-3}{2}\left[\lim_{z\to 0}\frac{-2}{(z-2)^3}\right]$$

$$= -\frac{3}{2}\left[\frac{-2}{-8}\right]$$

$$= -3/8$$

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Hence by Cauchy's Residue theorem

$$\oint_C \frac{(z+1)\,dz}{z^4-2z^3} = 2\pi i \left[\operatorname*{Res}_{z=0} f(z)\right]$$

$$= 2\pi i \operatorname*{Res}_{z=0} \left(\frac{z+4}{z^4-2z^3}\right)$$

$$= 2\pi i \times -\frac{3}{8}$$

$$= \frac{-3\pi i}{4}$$